Maclaurin Series & Convergence of Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the coefficient of x^2 in the Maclaurin series of e^{3x}.

Final Answer: $\frac{9}{2}$. $e^{3x} = \sum \frac{(3x)^k}{k!} \Rightarrow \text{coeff of } x^2 = \frac{3^2}{2!} = \frac{9}{2}$

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

Step 1: Test for absolute convergence. Consider $\sum |a_n| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$. This is a p-series with $p = 1/2 \le 1$, which diverges. Hence the series does NOT converge absolutely. Step 2: Apply Alternating Series Test. Here $b_n = \frac{1}{\sqrt{n}} > 0$. 1. Monotonicity: \(n+1 > n \implies \sqrt{n+1} > \sqrt{n} \implies \frac{1}{\sqrt{n+1}} conditionally convergent.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

Compute the limit of the general term as $n \to \infty$: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \cos\left(\frac{1}{n}\right) = \cos\left(\lim_{n\to\infty} \frac{1}{n}\right) = \cos(0) = 1$ Because $\lim_{n\to\infty} a_n = 1e0$, the series diverges by the Divergence Test.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x) = \cos x$.

Compute derivatives at $x = 0$: • $f(0) = \cos(0) = 1$ • $f'(0) = -\sin(0) = 0$ • $f''(0) = -\cos(0) = -1$ • $f'''(0) = \sin(0) = 0$ Applying the Maclaurin formula: $P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 = 1 - \frac{x^2}{2}$. (Notice the cubic term has coefficient 0).

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

6 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Find the first three non-zero terms of the Maclaurin series for e^{2x}. State the range of validity.

$e^{2x} = 1 + 2x + \frac{(2x)^2}{2!} + \cdots = 1 + 2x + 2x^2 + \cdots$. Valid for all $x \in \mathbb{R}$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

(a) Use $u_n = a + (n-1)d$: $u_3 = a + 2d = 14$ $u_7 = a + 6d = 34$ Subtract the first equation from the second: $4d = 20 \Rightarrow d = 5$. Substitute back: $a + 2(5) = 14 \Rightarrow a = 4$. (b) Use the sum formula $S_n = \frac{n}{2}(2a + (n-1)d)$ with $n = 25$: $S_{25} = \frac{25}{2}(2(4) + 24(5)) = \frac{25}{2}(8 + 120) = \frac{25}{2}(128) = 25 \times 64 = 1600$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.

Construct the Taylor polynomial using $a = 2$: $P_3(x) = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$ $= 3 - 1(x-2) + \frac{4}{2}(x-2)^2 + \frac{-12}{6}(x-2)^3 = 3 - (x-2) + 2(x-2)^2 - 2(x-2)^3$ Substitute $x = 2.1$, so $x - 2 = 0.1$: $P_3(2.1) = 3 - 0.1 + 2(0.1)^2 - 2(0.1)^3 = 3 - 0.1 + 2(0.01) - 2(0.001)$ $= 3 - 0.1 + 0.02 - 0.002 = 2.918$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · Foundations Find the exact sum of the infinite geometric series: $\sum_{n=1}^{\infty} 3\left(-\frac{2}{5}\right)^n$

Identify the first term and ratio: When $n = 1$, $a = 3\left(-\frac{2}{5}\right)^1 = -\frac{6}{5}$. The common ratio is $r = -\frac{2}{5}$. Since $|r| = \frac{2}{5} < 1$, the series converges. Using the sum formula $S = \frac{a}{1-r}$: $S = \frac{-6/5}{1-(-2/5)} = \frac{-6/5}{7/5} = -\frac{6}{7}$

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Series & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

Compute second derivative: $f(x) = x^{-1} \Rightarrow f'(x) = -x^{-2} \Rightarrow f''(x) = 2x^{-3} = \frac{2}{x^3}$. On $[1, 3]$, $|f''(x)| = \frac{2}{x^3}$ is decreasing, so its maximum occurs at $x = 1$: $K = \frac{2}{1^3} = 2$. Using the Trapezoidal error bound with $a = 1, b = 3$: $|E_T| \le \frac{K(b-a)^3}{12n^2} = \frac{2(3-1)^3}{12n^2} = \frac{2(8)}{12n^2} = \frac{16}{12n^2} = \frac{4}{3n^2}$. Require.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k-3) = 0$ has real roots.

For real roots, the discriminant $\Delta = b^2 - 4ac \ge 0$. Here $a = k$, $b = 4$, $c = k - 3$: $\Delta = 4^2 - 4(k)(k-3) = 16 - 4k^2 + 12k \ge 0$ Divide the entire inequality by -4, reversing the inequality sign: $k^2 - 3k - 4 \le 0$ Factor the quadratic: $(k-4)(k+1) \le 0$. The critical values are $k = -1$ and $k = 4$. Since the parabola opens upward, the product is ≤ 0 between the roots: $-1 \le k \le 4$. Note that if $k = 0$, the equation becomes the linear equation $4x - 3 = 0$, which has one r.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.02 The Poisson Distribution

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

Rewrite in index form: $y = 4x^{-1/2} + 2x$. At $x = 4$: $y = 4(4^{-1/2}) + 2(4) = 4\left(\frac{1}{2}\right) + 8 = 2 + 8 = 10$. Point of contact is $(4, 10)$. Differentiate: $\frac{dy}{dx} = 4\left(-\frac{1}{2}x^{-3/2}\right) + 2 = -2x^{-3/2} + 2 = -\frac{2}{x^{3/2}} + 2$ At $x = 4$: $x^{3/2} = (\sqrt{4})^3 = 8$. Thus: $m_{\tan} = -\frac{2}{8} + 2 = -\frac{1}{4} + 2 = \frac{7}{4}$ The normal has gradient $m_{\text{norm}} = -\frac{1}{7/4} = -\frac{4}{7}$. Using point $(4, 10)$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Further Integration Techniques

Level 2 · Medium Find the total arc length of the parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (the astroid) for $0 \le t \le \frac{\pi}{2}$.

Compute derivatives: $\frac{dx}{dt} = 3\cos^2 t(-\sin t) = -3\cos^2 t\sin t$ $\frac{dy}{dt} = 3\sin^2 t(\cos t) = 3\sin^2 t\cos t$ Form the integrand: $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\cos^4 t\sin^2 t + 9\sin^4 t\cos^2 t = 9\cos^2 t\sin^2 t(\cos^2 t + \sin^2 t) = 9\cos^2 t\sin^2 t$ Since $t \in [0, \pi/2]$, $\cos t \ge 0$ and $\sin t \ge 0$, so the square root is: $\sqrt{9\cos^2 t\sin^2 t} = 3\cos t\sin t = \frac{3}{2}\sin(2t)$ Integrate on $[0, \pi/2]$.

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17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.02 Differential Equations**

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

Substitute the double angle identity $\cos 2x = 1 - 2\sin^2 x$: $2(1 - 2\sin^2 x) = 1 + \sin x$ $2 - 4\sin^2 x = 1 + \sin x \Rightarrow 4\sin^2 x + \sin x - 1 = 0$ Apply quadratic formula for $\sin x$: $\sin x = \frac{-1 \pm \sqrt{1 - 4(4)(-1)}}{8} = \frac{-1 \pm \sqrt{17}}{8}$ Case 1: $\sin x = \frac{-1 + \sqrt{17}}{8} \approx 0.3904$: $x = 0.401 \text{ rad}, \ \pi - 0.401 = 2.74 \text{ rad}$ Case 2: $\sin x = \frac{-1 - \sqrt{17}}{8} \approx -0.6404$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)**

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

Here $f(x) = 2x$ and $f'(x) = 2$. The arc length element is $ds = \sqrt{1 + (f'(x))^2}\,dx = \sqrt{1 + 2^2}\,dx = \sqrt{5}\,dx$. The surface area formula gives: $S = 2\pi\int_0^3 f(x)\sqrt{1 + [f'(x)]^2}\,dx = 2\pi\int_0^3 (2x)\sqrt{5}\,dx = 4\sqrt{5}\pi\left[\frac{x^2}{2}\right]_0^3 = 4\sqrt{5}\pi\left(\frac{9}{2}\right) = 18\sqrt{5}\pi$ Verification via cone geometry: The shape is a right circular cone with radius $r = f(3) = 6$ and slant height $l = \sqrt{3^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$. Lateral are.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Matrices (AI HL)**

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

Take the natural logarithm of both sides: $\ln(3^{2x+1}) = \ln(5^{x-2})$. Apply the power law of logarithms: $(2x + 1)\ln 3 = (x - 2)\ln 5$. Distribute: $2x\ln 3 + \ln 3 = x\ln 5 - 2\ln 5$. Collect all terms containing x on one side: $2x\ln 3 - x\ln 5 = -\ln 3 - 2\ln 5 \Rightarrow x(2\ln 3 - \ln 5) = -(\ln 3 + 2\ln 5)$. Simplify logarithmic coefficients: $2\ln 3 = \ln 9$ and $2\ln 5 = \ln 25$: $x(\ln 9 - \ln 5) = -(\ln 3 + \ln 25) \Rightarrow x\ln\left(\frac{9}{5}\right) = -\ln(75)$. Solve for x.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)**

A block of mass $8\ \text{kg}$ rests in limiting equilibrium on a rough plane inclined at 30° to the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) calculate the normal reaction R, (b) find the coefficient of friction μ between the block and the plane.

(a) Resolve perpendicular to the inclined plane: $R = mg\cos 30^\circ = 8(9.8)\cos 30^\circ = 78.4 \times \frac{\sqrt{3}}{2} = 39.2\sqrt{3} \approx 67.9\ \text{N (3 s. f.)}$ (b) Resolve parallel to the plane. Since the block is in limiting equilibrium on the verge of sliding down, friction $F = \mu R$ acts up the plane: $F = mg\sin 30^\circ = 8(9.8)\sin 30^\circ = 78.4 \times 0.5 = 39.2\ \text{N}$ Setting $F = \mu R$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the general formula for the Maclaurin series expansion of a function $f(x)$?

a. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots$ ✓

b. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(0)x^n$

c. $f(x) = \sum_{n=1}^{\infty} \frac{f(n)}{n!}x^n$

d. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!}x^n$

A Maclaurin series is a Taylor series centered at $x = 0$: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the first three non-zero terms of the Maclaurin expansion of $f(x) = \cos(2x)$.

a. $1 - 2x^2 + \frac{2}{3}x^4$ ✓

b. $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

c. $1 - 4x^2 + 16x^4$

d. $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

Standard series $\cos u = 1 - \frac{u^2}{2} + \frac{u^4}{24} - \ldots$. Substitute $u = 2x$:
$1 - \frac{(2x)^2}{2} + \frac{(2x)^4}{24} = 1 - 2x^2 + \frac{16x^4}{24} = 1 - 2x^2 + \frac{2}{3}x^4$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the coefficient of x^3 in the Maclaurin series of e^{-3x}?

a. $-\frac{9}{2}$ ✓

b. $\frac{9}{2}$

c. -9

d. -27

$e^u = \sum \frac{u^n}{n!}$. For $n = 3$, $\frac{(-3x)^3}{3!} = \frac{-27x^3}{6} = -\frac{9}{2}x^3$. The coefficient is $-\frac{9}{2}$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the Maclaurin series for $\ln(1+3x)$ up to the x^3 term.

a. $3x - \frac{9}{2}x^2 + 9x^3$ ✓

b. $3x + \frac{9}{2}x^2 + 9x^3$

c. $x - \frac{1}{2}x^2 + \frac{1}{3}x^3$

d. $3x - 9x^2 + 27x^3$

Standard series $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$. With $u = 3x$: $(3x) - \frac{(3x)^2}{2} + \frac{(3x)^3}{3} = 3x - \frac{9}{2}x^2 + 9x^3$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate $\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$ using Maclaurin series.

a. $\frac{1}{2}$ ✓

b. 1

c. 0

d. ∞

$e^x = 1 + x + \frac{x^2}{2} + \dots$. Then $e^x - 1 - x = \frac{x^2}{2} + O(x^3)$. Dividing by x^2: $\frac{1}{2} + O(x) \to \frac{1}{2}$ as $x \to 0$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the first two non-zero terms of the series for $\frac{1}{1+x^2}$.

a. $1 - x^2$ ✓

b. $1 + x^2$

c. $1 - x$

d. $x - x^3$

Geometric series $\frac{1}{1-u} = 1 + u + u^2 + \dots$. With $u = -x^2$: $1 - x^2 + x^4 - \dots$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What does the Ratio Test state for a positive series $\sum a_n$ when $L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$?

a. Converges if $L < 1$, diverges if $L > 1$, test is inconclusive if $L = 1$ ✓

b. Converges if $L = 1$, diverges otherwise

c. Diverges if $L < 1$, converges if $L > 1$

d. Always inconclusive for power series

The Ratio Test establishes absolute convergence when the limiting ratio $L < 1$, divergence when $L > 1$, and provides no information when $L = 1$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Determine the convergence of $\sum_{n=1}^{\infty} \frac{n}{3^n}$ using the Ratio Test.

a. Converges, because $L = \lim \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} = \frac{1}{3} < 1$ ✓

b. Diverges, because $L = 3 > 1$

c. Inconclusive, because $L = 1$

d. Diverges, because the terms do not approach 0

$\lim_{n\to\infty} \frac{n+1}{n} \cdot \frac{3^n}{3^{n+1}} = 1 \cdot \frac{1}{3} = \frac{1}{3}$. Since $L = 1/3 < 1$, the series converges.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the limit L in the Ratio Test for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

a. 1 (test is inconclusive) ✓

b. 0

c. ∞

d. 1/2

$\lim_{n\to\infty} \frac{1/(n+1)}{1/n} = \lim \frac{n}{n+1} = 1$. The Ratio Test is inconclusive ($L = 1$); the harmonic series diverges by the p-series test ($p = 1$).

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$.

a. $R = \frac{1}{2}$ ✓

b. $R = 2$

c. $R = 1$

d. $R = \infty$

Ratio test: $\left|\frac{(2x)^{n+1}}{n+1} \cdot \frac{n}{(2x)^n}\right| = 2|x|\frac{n}{n+1} \to 2|x|$. For convergence: $2|x| < 1 \Rightarrow |x| < 1/2$. Thus $R = 1/2$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the radius of convergence of the Maclaurin series for e^x?

a. $R = \infty$ ✓

b. $R = 1$

c. $R = e$

d. $R = 0$

The exponential series converges for all complex numbers $x \in \mathbb{C}$; its radius of convergence is infinite ($R = \infty$).

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For $\sum_{n=0}^{\infty} x^n$, what is the exact interval of convergence?

a. $(-1, 1)$ ✓

b. $[-1, 1]$

c. $(-1, 1]$

d. $[-1, 1)$

Geometric series with ratio x: converges if and only if $|x| < 1$. At endpoints $x = \pm 1$, terms do not approach 0, so neither endpoint converges.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Reduction formula $I_n = \int_0^{\pi/2} \sin^n x\,dx$: $I_n = \frac{n-1}{n} I_{n-2}$. Find I_4:

a. $3\pi/8$

b. $3\pi/16$ ✓

c. $\pi/4$

d. $\pi/2$

I■=(3/4)I■=(3/4)(1/2)I■=(3/4)(1/2)(π/2)=3π/16.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The Maclaurin series for $\sqrt{1+x}$ is a special case of the binomial series with $\alpha =$

a. 2

b. 1

c. $-1/2$

d. $1/2$ ✓

Applying Convergence Tests and Power Series principles, $(1+x)^{1/2}$. Therefore, the verified correct option is $1/2$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The Maclaurin series for e^{-x^2} (substitute $-x^2$ for x in e^x series):

a. $\sum \frac{(-1)^n x^{2n}}{n!}$ ✓

b. $\sum x^{2n}/n!$

c. $\sum (-x)^n/n!$

d. $\sum x^n/n!$

$e^{-x^2} = \sum \frac{(-x^2)^n}{n!} = \sum \frac{(-1)^n x^{2n}}{n!}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Maclaurin series for $\tan^{-1} x$ at $x = 1$ gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$. This converges very slowly. The 10th partial sum approximates $\pi/4$ with error less than:

a. 1/11

b. $1/21$ ✓

c. 1/19

d. 0.01

Alternating series: error $\leq a_{11} = 1/21$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Maclaurin series for e^x: first four terms:

a. $1 + x + x^2 + x^3$

b. $1 + x + x^2/2! + x^3/3!$ ✓

c. $x + x^2 + x^3$

d. $1 - x + x^2 - x^3$

$e^x = \sum x^n/n! = 1 + x + x^2/2 + x^3/6 + \cdots$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R=3$ and the center of the series is $a=2$, which of the following intervals represents the values of x for which the series converges?

a. $(-1,5)$ ✓

b. $[-1,5]$

c. $(-1,5]$

d. $[-1,5)$

The radius of convergence R of a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ indicates that the series converges absolutely for $|x-a|R$. The center of the series is a. In this problem, we are given $R=3$ and $a=2$. Thus, the series converges for $|x-2|<3$. This inequality can be rewritten as $-3<x-2<3$. Adding 2 to all parts of the inequality, we get $-3+2<x<3+2$, which simplifies to $-1<x<5$. This interval is written in open interval notation as $(-1,5)$. The convergence at the endpoints.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Matrices (AI HL)

A sequence satisfies $u_{n+1}=2u_n+4$ with $u_1=3$. Find u_4.

a. 59

b. 33

c. **52** ✓

d. 39

Working through this additional-topics step: $u_2=2(3)+4=10,\ u_3=2(10)+4=24,\ u_4=2(24)+4=52$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A graph has vertices $\{1,\ldots,6\}$ and edges $1{-}5, 1{-}4, 4{-}5, 2{-}6, 2{-}5, 1{-}3$. Find the degree of vertex 2.

a. 0

b. 1

c. 3

d. **2** ✓

Vertex 2 appears in 2 edge(s), so its degree is 2.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.03 Markov Chains (AI HL)

A sequence is defined by $u_1 = 8$ and $u_{n+1} = u_n + 6$. Find u_4 given $u_3 = 20$.

a. **26** ✓

b. 34

c. 27

d. 32

Working through this additional-topics step: $u_4 = u_3 + 6 = 20 + 6 = 26$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Proof by Mathematical Induction

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = 0$.

a. 16°

b. 0° ✓

c. -34°

d. 18°

Working through this additional-topics step: $\sin^{-1}(0) = 0^\circ$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

A graph's adjacency matrix is $A = (0,1,1;1,0,1;1,1,0)$ (vertices 1,2,3). Find the number of walks of length 2 from vertex 1 to vertex 1 (the $(1,1)$ entry of A^2).

a. 1

b. 0

c. **2** ✓

d. 4

Computing A^2 gives entry $(1,1) = 2$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Find the future value of an ordinary annuity of $x500 deposited annually for 6 years at 2\%$ annual interest, correct to 2 decimal places.

a. \2,838.65

b. **\3,154.06** ✓

c. \3,469.47

d. \3,627.17

Working through this additional-topics step: $x3,154.06$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.04 Partial Fractions

Consider the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n \cdot 3^n}$. For what values of x does this series converge conditionally?

a. $-1 \leq x < 2$

b. $-1 < x \leq 2$ ✓

c. $-1 \leq x \leq 2$

d. $-1 < x < 2$

Let the given power series be $\sum_{n=1}^{\infty} a_n$, where $a_n = \frac{(2x-1)^n}{n \cdot 3^n}$. We will use the Ratio Test to find the interval of convergence. $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \to \infty} \left| \frac{(2x-1)^{n+1}}{(n+1) \cdot 3^{n+1}} \cdot \frac{n \cdot 3^n}{(2x-1)^n} \right|$.

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